

AIM-RGL: Agentic MI-CBT Framework with Reward-Guided Lookahead Search for Managing Client Resistance

by

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AUTHOR'S DECLARATION

I, Lu Htoo Kyaw, declare that the research work carried out for this thesis was in accordance with the regulations of the Asian Institute of Technology. The work presented in it are my own and has been generated by me as the result of my own original research, and if external sources were used, such sources have been cited. It is original and has not been submitted to any other institution to obtain another degree or qualification. This is a true copy of the thesis including final revisions.

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ABSTRACT

The growing demand for psychological support has increased interest in using large language models (LLMs) as accessible counseling agents. However, therapeutic dialogue requires more than generating a locally appropriate response. In counseling, especially when clients show resistance or ambivalence, the timing of therapeutic strategies is critical. Motivational Interviewing (MI) is useful for supporting readiness, autonomy, and exploration of ambivalence, while Cognitive Behavioral Therapy (CBT) provides structured techniques for identifying and modifying unhelpful thoughts and behaviors. Introducing CBT too early may increase resistance, while remaining only in MI may slow progress when the client is ready for structured intervention.

Existing LLM-based counseling systems have explored therapeutic strategy selection, memory-based adaptation, and MI-CBT dialogue support. However, many approaches remain limited in their ability to anticipate the immediate downstream effects of a therapeutic response before it is selected. As a result, a system may choose a response that appears suitable in the current turn but does not support the client’s next therapeutic state, readiness, or engagement.

This thesis proposes AIM-RGL, an agentic MI-CBT framework with reward-guided lookahead search for managing client resistance in simulated therapeutic conversations. The framework uses multiple specialist therapist agents, including open-ended questioning, reflection, affirmation, summarization, and CBT agents, to generate candidate responses. A client simulator then produces possible future client reactions based on cognitive conceptualization and stage-of-change information. A reward model evaluates candidate responses according to therapeutic dimensions such as alliance, empathy, collaboration, CBT skill, MI quality, and client readiness. Instead of selecting only the best immediate response, AIM-RGL performs short-horizon lookahead search to choose the response expected to lead to a better near-future therapeutic state.

The study evaluates whether reward-guided lookahead improves MI-CBT response quality compared with baseline approaches. It further examines how different lookahead depths, reward model backbones, and policy model backbones affect therapeutic response generation. Evaluation focuses on simulated therapist-client dialogues using metrics related to therapeutic alliance, MI fidelity, CBT competence, change talk, and

sustain talk. This thesis does not claim to measure real clinical treatment outcomes or replace professional therapists; rather, it investigates how future-aware response selection can improve the design of LLM-based counseling support systems. Overall, AIM-RGL contributes a structured framework for proactive, context-sensitive, and therapeutically grounded response generation in AI-assisted mental health dialogue.

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Mental health support has become an increasingly important global concern as psychological distress, anxiety, depression, and related conditions continue to affect people's daily functioning, relationships, work, and quality of life. The World Health Organization reported that more than one billion people are living with mental health disorders, with anxiety and depression among the most common conditions worldwide World Health Organization (2025). Despite increasing awareness of mental health issues, access to professional psychological care remains limited, particularly in low- and middle-income countries where treatment gaps are still high Kazdin (2019); World Health Organization (2025). These challenges have encouraged researchers to explore scalable forms of psychological support, including digital mental health interventions and large language model-based counseling systems. Recent studies suggest that large language models may support mental health-related tasks, but their use in counseling requires careful evaluation because therapeutic dialogue involves empathy, safety, contextual understanding, and professional boundaries Hua, Na, Li, et al. (2025); Nguyen et al. (2025).

Large language models have shown potential for producing fluent and empathetic responses that fit counseling-like contexts. However, therapeutic communication is not only a matter of generating a warm or natural reply. A helpful response also needs to preserve trust, respect the client's autonomy, and support a collaborative relationship between the client and the therapist. These qualities are central to client-centered counseling and therapeutic alliance, where empathy, agreement on goals, and agreement on therapeutic tasks influence the quality of the interaction Bordin (1979); Rogers (1957). For LLM-based counseling systems, this means that response generation should be grounded not only in language quality, but also in therapeutic principles and interactional sensitivity.

This thesis focuses on simulated therapeutic dialogue involving resistance and ambivalence. Motivational Interviewing (MI) and Cognitive Behavioral Therapy (CBT) provide two complementary foundations for this setting. MI emphasizes collaboration, autonomy, reflective listening, affirmation, and eliciting the client's own reasons for

change, making it useful when the client is uncertain or not yet ready to engage deeply with change Bischof, Bischof, and Rumpf (2021); Miller and Rollnick (2013). CBT provides structured methods for identifying unhelpful thoughts, examining evidence, developing more balanced interpretations, and supporting behavioral change Beck (2020); Chand, Kuckel, and Huecker (2023). Together, MI and CBT offer a useful combination for supporting clients who may move between hesitation, exploration, and readiness for more structured therapeutic work.

The integration of MI and CBT has also been explored in conventional mental health treatment outside the LLM context. Arkowitz and Westra discussed the use of MI with CBT for depression and anxiety, especially where resistance and ambivalence affect treatment engagement Arkowitz and Westra (2004). In generalized anxiety disorder, studies have examined MI as a pretreatment or integrated component of CBT, with findings suggesting potential benefits for clients with higher severity, resistance, or ambivalence Westra, Arkowitz, and Dozois (2009); Westra, Constantino, and Antony (2016). Reviews of MI as an adjunct to CBT for anxiety disorders also suggest that MI-CBT can support treatment initiation, engagement, and clinical outcomes, although the literature still points to the importance of how and when MI is integrated with CBT Marker and Norton (2018); Randall and McNeil (2017). This clinical background supports the motivation of this thesis: the challenge is not only combining MI and CBT, but coordinating them at the right moment in the therapeutic dialogue.

The importance of timing in therapeutic dialogue can be understood through the client's readiness for change. The Transtheoretical Model describes behavior change as a staged process that includes precontemplation, contemplation, preparation, action, and maintenance Prochaska and Velicer (1997); Raihan and Cogburn (2023). A client in precontemplation may minimize the problem or resist direct analysis, while a client in preparation may be more open to structured reflection, cognitive restructuring, or planning. Therefore, the same therapeutic move may be helpful in one stage but ineffective or even counterproductive in another. In this thesis, resistance and readiness are treated as important interaction states that shape whether an MI-style or CBT-style response is more suitable.

These considerations motivate the need for LLM-based counseling systems that are sensitive to the possible direction of the dialogue, not only to the current user message. In

therapeutic conversation, a response should ideally help the client move toward greater engagement, openness, collaboration, or readiness. Short-horizon lookahead provides one way to think about this problem, because it allows multiple candidate responses and possible future client reactions to be considered before selecting the final response. Similar search-based ideas have been explored in large language model reasoning, where considering multiple candidate paths can improve decision-making in complex tasks Yao et al. (2023). This thesis adapts this general idea to simulated MI-CBT therapeutic dialogue by investigating whether future-aware response selection can support better near-future therapeutic states.

To address this direction, this thesis proposes AIM-RGL, an agentic MI-CBT framework with reward-guided lookahead search for managing client resistance in simulated therapeutic conversations. The framework uses specialist therapist agents to generate candidate responses, including agents for open-ended questioning, reflection, affirmation, summarization, and CBT. A client simulator then produces possible future client reactions based on cognitive conceptualization and stage-of-change information, while a reward model evaluates candidate trajectories using therapeutic dimensions such as alliance, empathy, collaboration, CBT skill, MI quality, and client readiness. Instead of selecting only a locally appropriate response, AIM-RGL aims to select the response expected to support a better near-future therapeutic state. This thesis does not claim to diagnose, treat, replace professional therapists, or evaluate real clinical outcomes; rather, it investigates future-aware response selection in simulated AI-assisted counseling dialogue.

1.2 Statement of the Problem

Although LLM-based counseling systems have become increasingly capable of generating fluent and empathetic responses, many still treat therapeutic response generation mainly as an immediate-context decision. This creates a problem because therapeutic response quality depends not only on how appropriate a response appears in the present turn, but also on how it affects the client’s next state. A response may sound supportive but still increase defensiveness, reduce engagement, or weaken the therapeutic alliance if it does not match the client’s readiness for change Bordin (1979); Rogers (1957). Recent evaluations of LLM-based counseling systems also highlight the need to assess these systems using counseling-specific competencies, safety, and therapeutic quality rather than

relying only on general language quality Hua et al. (2025); Nguyen et al. (2025). Therefore, the central problem addressed in this thesis is the lack of future-aware therapeutic response selection in simulated LLM-based counseling dialogue.

This problem is especially important when an LLM-based system needs to coordinate MI and CBT responses for resistant or ambivalent clients. MI and CBT support different therapeutic needs: MI helps the client explore ambivalence and motivation, while CBT supports structured examination of thoughts and behavior change Beck (2020); Bischof et al. (2021); Chand et al. (2023); Miller and Rollnick (2013). However, the usefulness of each approach depends on when it is applied. If CBT is introduced before the client is ready, it may feel directive and increase resistance; if the system remains only in MI after the client becomes ready, it may miss opportunities for structured therapeutic progress Prochaska and Velicer (1997); Raihan and Cogburn (2023). Therefore, the problem is not simply choosing between MI and CBT, but selecting the therapeutic move that best fits both the client’s current state and likely near-future response.

Prior work on LLM-based counseling has studied memory, stage awareness, strategy selection, and therapeutic response evaluation. For example, ChatThero uses a multi-session, stressor-aware language agent for recovery support, while Persona Drift explores adaptive flow in stage-aware CBT chatbots J. Wang et al. (2025); Yun, Kim, Kim, and Oh (2025). More closely related to this thesis, FLASH proposes a hybrid MI-CBT counseling framework that uses Flashbulb Memory to store high-impact client reactions to specific strategies and dynamically guide later intervention selection based on historical success and failure Anonymous (2026). However, these systems mainly adapt through memory, stage awareness, or previously observed interaction signals, and may still have limited ability to compare the immediate downstream effects of multiple candidate therapeutic moves before selecting a final response. This creates a gap between responses that fit the current turn and decisions that consider future dialogue states. This thesis addresses the gap by testing whether reward-guided lookahead search can compare multiple candidate MI-CBT responses and select the response expected to produce a better simulated client trajectory.

1.3 Objectives

The main objective of this thesis is to develop and evaluate AIM-RGL, an agentic MI-CBT framework that uses reward-guided lookahead search to improve therapeutic re-

sponse selection for managing client resistance in simulated counseling dialogue.

- To build a multi-agent therapist architecture that generates candidate responses with specialist agents for open-ended questioning, reflection, affirmation, summarization, and CBT.
- To design a simulated client interaction setting grounded in cognitive conceptualization and stage-of-change information, so that client resistance, ambivalence, and readiness can be modeled during therapeutic dialogue.
- To train a reward model that estimates the therapeutic value of candidate responses based on dimensions such as therapeutic alliance, empathy, collaboration, CBT skill, MI quality, and client readiness.
- To implement reward-guided lookahead search that compares multiple candidate MI-CBT responses and selects the response expected to support a better near-future client trajectory.
- To examine how different lookahead horizons, including one-step, two-step, and three-step lookahead, affect the quality of MI-CBT therapeutic response selection.
- To evaluate whether AIM-RGL improves therapeutic skill quality, MI-CBT fidelity, therapeutic alliance, change talk, and sustain talk compared with selected baseline counseling systems.
- To analyze the effect of different reward model backbones and policy model backbones on the performance of reward-guided MI-CBT response generation.

1.4 Organization of the Study

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CHAPTER 2

LITERATURE REVIEW

This chapter reviews the main theoretical and technical foundations related to this thesis. It first discusses therapeutic communication, client resistance, readiness for change, Motivational Interviewing, and Cognitive Behavioral Therapy as the clinical basis for MI-CBT response selection. It then reviews prior LLM-based counseling systems, including approaches that use memory, stage awareness, strategy selection, and therapeutic evaluation. Finally, the chapter connects these works to search-based and reward-guided methods, highlighting the gap that motivates AIM-RGL: the need for a system that can compare possible near-future client reactions before selecting a therapeutic response.

2.1 Clinical Foundations of MI-CBT Therapeutic Dialogue

Therapeutic dialogue differs from ordinary conversation because the therapist's response should support understanding, collaboration, and meaningful change. A counseling response is not evaluated only by relevance or fluency, but also by whether it strengthens the therapeutic relationship, respects client autonomy, and fits the client's current psychological state. This is especially important when the client shows resistance, ambivalence, or low readiness for change, because poorly timed responses may reduce engagement or increase defensiveness Beutler, Moleiro, and Talebi (2002); Patterson and Chamberlain (1994); Patterson and Forgatch (1985). Figure 2.1 illustrates how client state, response timing, therapeutic outcomes, and the next client state are connected.

2.1.1 Therapeutic Alliance and Client-Centered Communication

Therapeutic alliance refers to the collaborative relationship between the therapist and the client. Bordin described the working alliance as involving agreement on goals, agreement on therapeutic tasks, and the emotional bond between therapist and client Bordin (1979). Rogers also emphasized empathy, genuineness, and unconditional positive regard as core conditions for therapeutic personality change Rogers (1957). These foundations suggest that therapeutic dialogue depends not only on the intervention itself, but also on how the intervention is communicated.

Client-centered communication is important in AI-assisted counseling because an LLM may generate a response that sounds supportive but still feels directive, assumptive, or

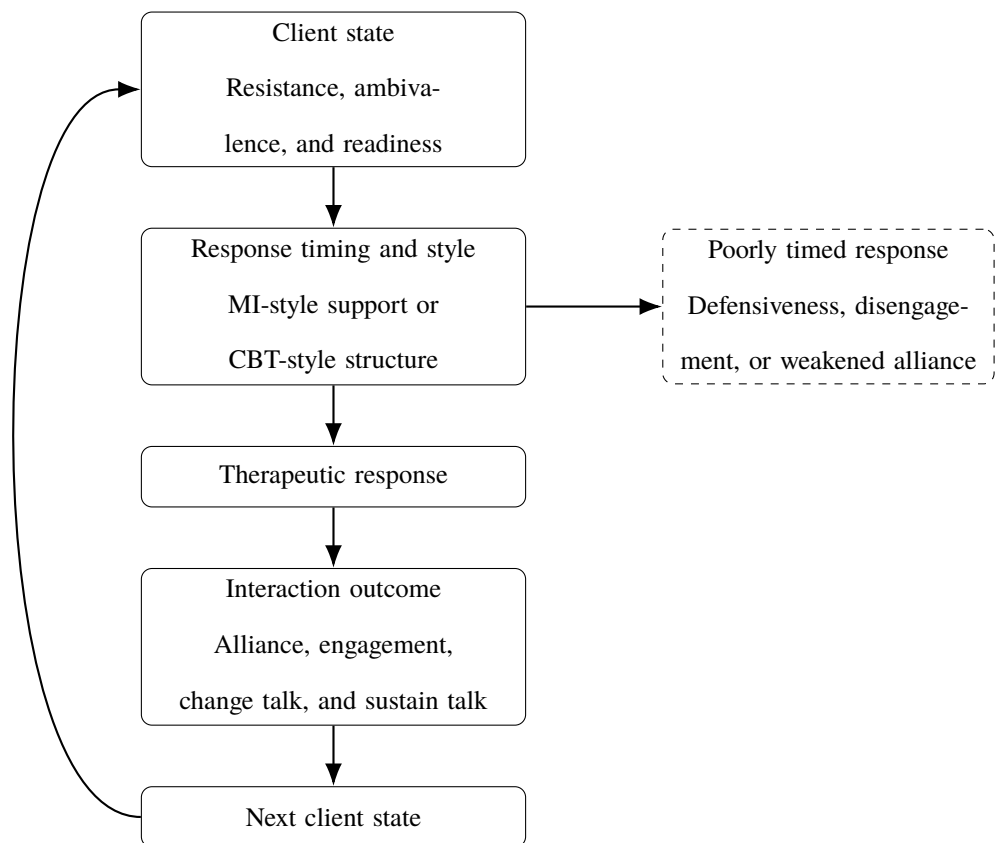


Figure 2.1

Conceptual relationship between client state, response timing, therapeutic outcomes, and the next client state in therapeutic dialogue.

poorly matched to the client's readiness. For example, a response may appear warm while still pushing the client toward action too quickly. In MI-CBT dialogue, this is important because structured CBT techniques may require sufficient trust and collaboration before they are introduced. Therefore, therapeutic alliance is treated as a key quality dimension in this thesis when evaluating whether a response supports engagement and readiness for further therapeutic work.

2.1.2 Client Resistance, Ambivalence, and Readiness for Change

Client resistance refers to reluctance, opposition, or disengagement during the therapeutic process. It may appear openly through arguing or rejecting suggestions, or more subtly through short answers, avoidance, topic changes, or passive agreement Patterson and Chamberlain (1994); Patterson and Forgatch (1985). Resistance has also been discussed as a process that may be influenced by therapist behavior, rather than only as a fixed client trait Beutler et al. (2002). In MI, resistance is often understood as a signal that the therapeutic approach may not fit the client's current readiness or concerns Miller and Rollnick (2013).

Ambivalence occurs when clients hold both reasons for and reasons against change. Motivational Interviewing addresses this tension by helping clients explore ambivalence and strengthen their own reasons for change Bischof et al. (2021); Miller and Rollnick (2013). Therapist responses can also influence whether clients produce more change talk or sustain talk during MI sessions Moyers et al. (2007). In this thesis, ambivalence is important because it creates moments where the system must decide whether the client needs further motivational support or is ready for structured CBT intervention.

Readiness for change explains why the same therapeutic response may help one client but hinder another. The Transtheoretical Model describes change as a staged process that includes precontemplation, contemplation, preparation, action, and maintenance Prochaska and Velicer (1997); Raihan and Cogburn (2023). A client in precontemplation may not yet view the issue as a problem, while a client in preparation may be more open to concrete strategies or behavioral steps. Therefore, therapeutic response selection should be sensitive to the client's current readiness rather than applying the same intervention style across all moments.

2.1.3 Motivational Interviewing, Cognitive Behavioral Therapy, and Their Integration

Motivational Interviewing is a counseling approach that supports motivation, autonomy, and readiness for change. It emphasizes collaboration, evocation, acceptance, and compassion, and commonly uses open-ended questions, affirmations, reflections, and summaries Bischof et al. (2021); Miller and Rollnick (2013). MI is especially useful when the client is ambivalent or resistant because it avoids direct confrontation and helps the client explore their own reasons for change. In this thesis, MI provides the foundation for response styles that reduce defensiveness and support readiness.

Cognitive Behavioral Therapy provides a more structured approach for identifying and modifying unhelpful thoughts, beliefs, and behaviors. CBT includes techniques such as identifying automatic thoughts, examining evidence, generating alternative interpretations, and planning behavioral change Beck (2020); Chand et al. (2023). These techniques are useful when the client is ready to reflect and take concrete steps, but they may feel directive if introduced too early. Therefore, CBT is most suitable when the client shows sufficient openness and the therapeutic relationship is stable.

The integration of MI and CBT has been explored in conventional mental health treatment, especially for clients who experience ambivalence or resistance during CBT. Prior work suggests that MI can support treatment engagement and prepare clients for more structured CBT work Arkowitz and Westra (2004); Westra et al. (2009, 2016). Reviews also indicate that MI may be useful as an adjunct to CBT, although the timing and manner of integration remain important Marker and Norton (2018); Randall and McNeil (2017). Table 2.1 summarizes the main differences between MI and CBT and their relevance to AIM-RGL.

Overall, these clinical foundations show that effective MI-CBT dialogue depends on resistance, ambivalence, readiness, and therapeutic alliance. MI and CBT offer complementary response styles, but their usefulness depends heavily on timing. This motivates AIM-RGL's focus on selecting responses not only by current-turn suitability, but also by their likely near-future effect on the simulated client's therapeutic state.

Table 2.1*Comparison between MI and CBT.*

Aspect	MI	CBT	Relevance to AIM-RGL
Purpose	Builds motivation, autonomy, and readiness.	Supports thought and behavior change.	Helps choose between motivational support and structured intervention.
Stance	Collaborative and autonomy-supportive.	Structured and skill-focused.	Response style should match resistance and readiness.
Best suited for	Ambivalence, resistance, or uncertainty.	Readiness for reflection, belief testing, or action.	Supports stage-sensitive response selection.
Risk if mis-timed	May slow structured progress if overused.	May feel directive if introduced too early.	Motivates future-aware MI-CBT timing.

2.2 LLM-Based Counseling Systems and Therapeutic Evaluation

LLM-based counseling systems have recently been explored as tools for generating supportive, empathetic, and therapy-informed dialogue. However, counseling dialogue requires more than fluent response generation because the system must also consider therapeutic safety, client readiness, alliance, and the appropriateness of the intervention strategy Hua et al. (2025); Nguyen et al. (2025). For this thesis, prior systems are reviewed in terms of how they support CBT, MI, hybrid MI-CBT interaction, client simulation, and therapeutic evaluation.

2.2.1 LLM-Based CBT, MI, and Hybrid Counseling Agents

Existing LLM-based counseling systems have explored different ways of supporting therapy-informed dialogue, including CBT-style response generation, MI-based behavior-change support, counseling dialogue datasets, client simulation, and hybrid MI-CBT interaction. CBT-oriented works usually focus on structured therapeutic skills such as identifying thoughts, cognitive reframing, guided discovery, or generating multi-turn CBT-style counseling conversations. For example, CACTUS constructs multi-turn counseling dialogues grounded in CBT and client personas, while HealMe focuses on

cognitive reframing through psychotherapy-informed dialogue Lee et al. (2024); Xiao et al. (2024). Psych8k and ChatCounselor further show the value of domain-specific counseling conversations for training mental health support models, as they use real counseling conversations to improve counseling-related response quality Liu et al. (2023).

MI-based and behavior-change systems focus more on motivation, autonomy, readiness, and the client's movement toward change. CAMI is an MI-grounded counselor agent that uses state inference, topic exploration, action selection, and response generation to support behavior-change counseling Yang, Achananuparp, Huang, Jiang, Kit, et al. (2025). CCSMI extends this direction by focusing on consistent client simulation for MI-based counseling, where the simulated client's mental state, motivation, beliefs, plans, and receptivity are tracked across the conversation Yang, Achananuparp, Huang, Jiang, Lim, et al. (2025). These works are relevant to this thesis because they show how LLM-based counseling systems can represent client state and readiness during multi-turn interaction.

Other studies focus on constructing or reconstructing counseling dialogues for training and evaluation. SMILE converts single-turn mental health support data into multi-turn conversations, while CPsyCoun reconstructs Chinese psychological counseling dialogues from counseling reports and proposes an evaluation framework for multi-turn counseling Qiu, He, Zhang, Li, and Lan (2024); C. Zhang et al. (2024). PATIENT- Ψ is also relevant because it uses CBT-based cognitive models to simulate patients for training mental health professionals, making it closely related to cognitive conceptualization and simulated client design R. Wang et al. (2024). MAGNeT further extends synthetic counseling data generation by using coordinated specialist agents to generate multi-turn mental health counseling sessions Mandal, Chakraborty, and Gurevych (2025).

More recent work has also explored adaptive and hybrid counseling agents. Persona Drift studies stage-aware adaptation in CBT chatbots, while ChatThero explores multi-session LLM-supported therapeutic support for behavior change J. Wang et al. (2025); Yun et al. (2025). FLASH is especially close to this thesis because it combines MI and CBT and uses Flashbulb Memory to guide strategy selection for managing client resistance Anonymous (2026). However, these systems mainly rely on current-state signals, stage adaptation, synthetic dialogue construction, or memory from previous interactions. Less attention has been given to explicitly comparing possible near-future client reactions before selecting a therapeutic response, which motivates the future-aware response

selection direction of this thesis.

Table 2.2 summarizes the LLM-based systems most closely related to MI-CBT coordination and adaptive therapeutic strategy selection.

2.2.2 Client Simulation and Cognitive Conceptualization

Client simulation is useful for evaluating LLM-based counseling systems before involving real users, especially in sensitive mental health-related settings. In this thesis, simulated clients provide controlled interaction scenarios involving resistance, ambivalence, and different levels of readiness. Cognitive conceptualization can support this simulation by representing the client's situation, beliefs, emotions, automatic thoughts, coping patterns, and behavioral responses. This allows the simulated client to respond more consistently across turns instead of producing isolated or random replies.

Figure 2.2 illustrates the general simulation loop used to evaluate therapeutic dialogue. The therapist-like system generates a response based on the conversation history, the simulated client produces a reaction based on the client profile and current state, and the resulting dialogue is evaluated using therapeutic metrics.

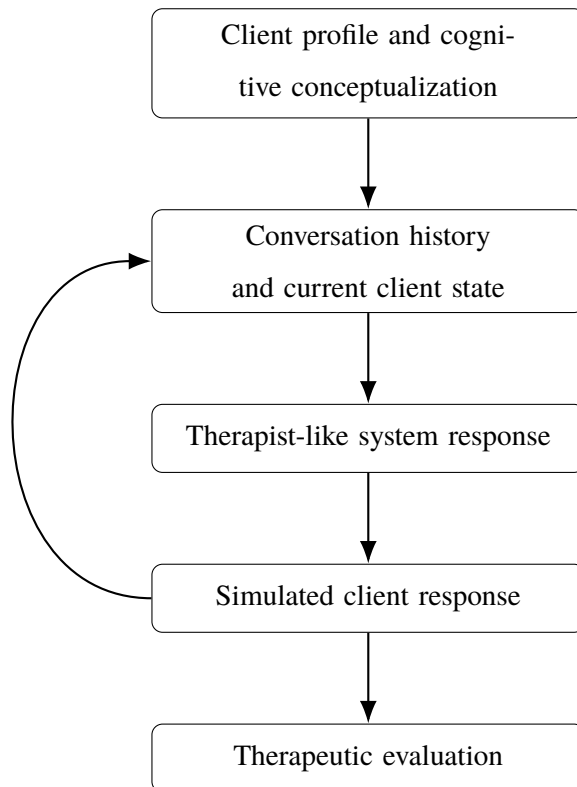


Figure 2.2

General client simulation loop for therapeutic dialogue evaluation.

Table 2.2

LLM-based systems related to MI-CBT coordination and adaptive therapeutic response selection.

System	Therapeutic Focus	Main Mechanism	Limitation / Gap
ChatThero J. Wang et al. (2025)	Behavior change with MI and CBT strategies	Uses a multi-session, stressor-aware, memory-persistent language agent for addiction recovery support.	Uses adaptive thera- peutic strategies, but is less focused on explicitly comparing short-horizon future client reactions before response selection.
Persona Drift Yun et al. (2025)	Stage-aware CBT adaptation	Adjusts CBT chatbot flow when persona or stage drift is detected.	Relevant to stage- aware adaptation, but it is not a full MI-CBT coordina- tion framework and does not directly compare future client trajectories.
FLASH Anony- mous (2026)	Hybrid MI-CBT for resistance manage- ment	Uses readiness esti- mation and Flashbulb Memory to dynami- cally guide switching between MI and CBT strategies.	Adapts mainly from observed client his- tory and accumulated memory, rather than explicitly comparing multiple simulated near-future client reac- tions before response selection.
AIM-RGL	Hybrid MI-CBT with reward-guided lookahead	Compares candidate MI-CBT responses and simulated near- future client reactions before selecting a response.	Proposed framework evaluated in simulated counseling dialogue.

2.2.3 Evaluation Metrics for Therapeutic Dialogue

Therapeutic dialogue evaluation should consider counseling-specific qualities rather than only general language quality. In this thesis, evaluation focuses on whether responses support CBT competence, MI consistency, therapeutic alliance, and movement toward change. CTRS is used to assess CBT-related therapist competence, MITI is used to assess MI fidelity, WAI is used to assess therapeutic alliance, and change talk and sustain talk are used to examine whether the simulated client's language moves toward or away from change Horvath and Greenberg (1989); Miller and Rollnick (2013); Miller and Rose (2009); Moyers, Rowell, Manuel, Ernst, and Houck (2016); Young and Beck (1980). These metrics are relevant because AIM-RGL aims to improve not only response fluency, but also therapeutic timing, collaboration, and simulated client readiness.

Because therapeutic alliance is central to this thesis, the alliance dimension is further represented through three components: goal agreement, approach or task agreement, and affective bond. These dimensions reflect whether the client and counselor share an understanding of therapy goals, agree on how therapeutic work should proceed, and maintain trust and emotional connection Bordin (1979); Hatcher and Gillaspay (2006); Horvath and Greenberg (1989). Table 2.4 summarizes the alliance-related questionnaire items used to evaluate these dimensions.

In addition to rating-based metrics, change talk and sustain talk can be represented as proportional indicators of the simulated client's movement toward or away from change. The proportions are defined as follows:

$$\text{Change Talk Ratio} = \frac{\text{Number of change talk utterances}}{\text{Total number of client utterances}} \quad (2.1)$$

$$\text{Sustain Talk Ratio} = \frac{\text{Number of sustain talk utterances}}{\text{Total number of client utterances}} \quad (2.2)$$

Together, these evaluation dimensions allow the study to assess whether AIM-RGL improves therapeutic response quality in terms of CBT skill, MI consistency, therapeutic alliance, readiness, and simulated client movement across the dialogue.

Table 2.3*Evaluation metrics for therapeutic dialogue.*

Metric	Source / Basis	Relevance to This Thesis
CTRS	Cognitive Therapy Rating Scale for evaluating CBT therapist competence Beck Institute for Cognitive Behavior Therapy (2021); Young and Beck (1980).	Evaluates CBT-style response quality, including structure, guided discovery, and cognitive intervention.
MITI	Motivational Interviewing Treatment Integrity code for assessing MI fidelity Moyers et al. (2015, 2016).	Evaluates whether responses follow MI principles such as empathy, autonomy support, and MI-consistent behavior.
WAI / Alliance	Working Alliance Inventory for measuring therapeutic alliance Hatcher and Gillaspay (2006); Horvath and Greenberg (1989).	Evaluates trust, collaboration, goal agreement, and task or approach agreement.
Change Talk	MI construct describing client language favoring change Miller and Rollnick (2013); Miller and Rose (2009).	Indicates movement toward readiness, motivation, or action.
Sustain Talk	MI construct describing client language favoring the current state Miller and Rollnick (2013); Miller and Rose (2009).	Indicates resistance, ambivalence, or lower readiness for change.

Table 2.4*Therapeutic alliance dimensions and questionnaire items.*

Dimension	Question	No.
Goal	There is mutual understanding about what participants are trying to accomplish in therapy.	Q1
	The client and counselor are working on mutually agreed upon goals.	Q2
	The client and counselor have the same ideas about the client's real problems.	Q3
	The client and counselor have established a good understanding of the changes that would be good for the client.	Q4
Approach	There is agreement about the steps taken to help improve the client's situation.	Q5
	There is agreement about the usefulness of the current activity in therapy.	Q6
	There is agreement on what is important for the client to work on.	Q7
	The client believes that the way they are working with the problem is correct.	Q8
Affective Bond	There is a mutual liking between the client and counselor.	Q9
	The client feels confident in the counselor's ability to help the client.	Q10
	The client feels that the counselor appreciates him/her as a person.	Q11
	There is mutual trust between the client and counselor.	Q12

2.3 Future-Aware Response Selection and Research Gap

Future-aware response selection refers to choosing an action by considering not only its immediate suitability, but also its possible downstream effects. This idea is common in artificial intelligence, especially in sequential decision-making problems where a current decision changes the quality of later states Sutton and Barto (2018). In game playing, mathematical reasoning, instruction following, and text generation, systems often generate multiple candidate paths, evaluate them, and select the option expected to lead to a better outcome. These methods are relevant to this thesis because therapeutic dialogue is also sequential: a response that appears appropriate in the current turn may still lead to resistance, disengagement, or weaker alliance in the next turn.

2.3.1 *Search-Based Reasoning and Lookahead Methods*

Search-based methods have been successful in domains where decisions require planning over future possibilities. In game playing, AlphaGo combined deep neural networks with tree search to evaluate possible future moves before selecting an action, showing that learned evaluation and search can support strong sequential decision-making Silver et al. (2016). In language-model reasoning, Self-Consistency improves chain-of-thought reasoning by sampling multiple reasoning paths and selecting the most consistent answer rather than relying on a single greedy path X. Wang et al. (2022). Tree of Thoughts further extends this idea by allowing language models to explore, evaluate, and backtrack across multiple intermediate reasoning paths Yao et al. (2023). These works show that exploring alternatives before committing to a final output can improve performance in tasks that require planning or reasoning.

More recent work combines search with reward signals during LLM inference or self-training. Training Verifiers for math word problems generates multiple candidate solutions and selects the one judged most likely to be correct by a verifier model Cobbe et al. (2021). ReST-MCTS* uses process-reward-guided Monte Carlo Tree Search to collect higher-quality reasoning traces and improve policy and reward models through self-training D. Zhang et al. (2024). Instead of relying only on final-answer correctness, these approaches evaluate intermediate steps or candidate paths. This suggests that future-aware selection is useful when the system must compare several possible paths before committing to one output.

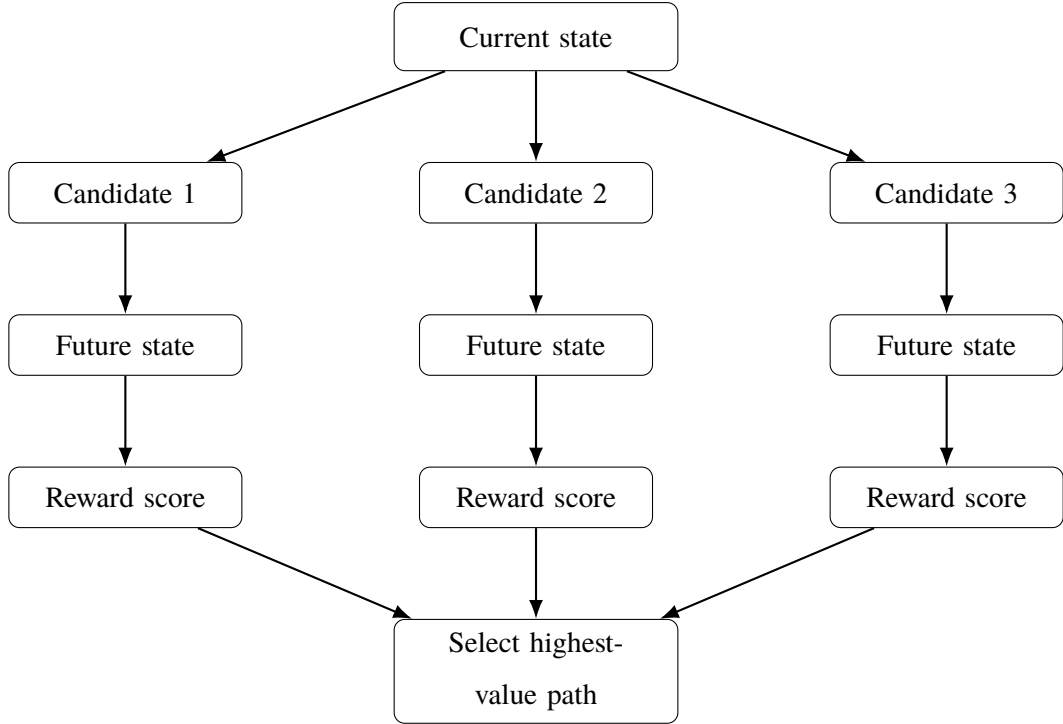


Figure 2.3

Generic reward-guided lookahead search for response selection.

2.3.2 Reward Models and Reward-Guided Selection

Reward models provide a way to convert human preferences, task success, or quality judgments into a score that can guide selection or training. In reinforcement learning from human preferences, Christiano et al. trained reward models from human comparisons of trajectory segments and used them to guide agents in Atari and simulated robotics tasks Christiano et al. (2017). In LLM alignment, InstructGPT trained a reward model from human rankings of model outputs and used reinforcement learning from human feedback to improve instruction-following behavior Ouyang et al. (2022). Direct Preference Optimization later showed that preference learning can also be optimized more directly without a separate reinforcement learning stage Rafailov et al. (2023). These works show that human or task-based preference signals can guide model behavior beyond simple next-token prediction.

Reward-guided selection has also been used directly for choosing among generated outputs. LLM-Blender uses PairRanker to compare candidate responses and GenFuser to combine high-ranked outputs, showing that ranking and selection can improve generation quality across multiple LLMs Jiang, Ren, and Lin (2023). In mathematical reasoning, process supervision trains reward models to evaluate intermediate reasoning steps

rather than only final answers, and has been shown to outperform outcome-only supervision on challenging reasoning tasks Lightman et al. (2023). These methods support the idea that response quality can be improved by evaluating candidate outputs or intermediate paths before producing a final answer.

Following reward-model and ranking-based response selection approaches, candidate outputs can be scored and the highest-valued response can be selected Cobbe et al. (2021); Jiang et al. (2023); Ouyang et al. (2022):

$$r_t^* = \arg \max_{r_t^{(i)} \in \mathcal{R}_t} \hat{R}(H_t, r_t^{(i)}) \quad (2.3)$$

where H_t is the current context, $\mathcal{R}_t$ is the set of candidate responses, $r_t^{(i)}$ is the i -th candidate response, $\hat{R}$ is the estimated reward function, and r_t^* is the selected response.

When future outcomes are considered, the value of a candidate trajectory can be represented using the standard discounted return formulation from reinforcement learning Sutton and Barto (2018):

$$G_t^{(i)} = \sum_{k=0}^{d-1} \gamma^k \hat{R}_{t+k}^{(i)} \quad (2.4)$$

where $G_t^{(i)}$ is the estimated value of candidate trajectory i , d is the lookahead depth, γ is the discount factor, and $\hat{R}_{t+k}^{(i)}$ is the predicted reward at future step k .

2.3.3 Research Gap and Positioning of AIM-RGL

The reviewed methods show that search, lookahead, ranking, and reward modeling can improve decision-making in several domains. In games, search helps evaluate future consequences before selecting a move. In reasoning tasks, lookahead allows models to compare multiple paths instead of committing to the first generated solution. In alignment and response generation, reward models and rankers help select outputs that better match human preferences or task goals.

However, these methods have mostly been applied to domains such as game playing, mathematical reasoning, instruction following, and general response generation. In LLM-based counseling, prior systems often focus on generating an appropriate current

Table 2.5*Search and reward-guided methods relevant to future-aware response selection.*

Study	Area	Method	Reported Contribution
AlphaGo Silver et al. (2016)	Game playing	Combines deep neural networks with tree search.	Shows that learned evaluation plus search can support strong sequential decision-making.
Self-Consistency X. Wang et al. (2022)	LLM reasoning	Samples multiple reasoning paths and selects the most consistent answer.	Improves reasoning by avoiding reliance on a single greedy path.
Tree of Thoughts Yao et al. (2023)	LLM reasoning	Explores and evaluates multiple intermediate reasoning paths.	Improves performance on tasks requiring planning, search, and backtracking.
Training Verifiers Cobbe et al. (2021)	Mathematical reasoning	Generates many solutions and selects the highest-ranked one using a verifier.	Shows that candidate generation plus learned verification can improve reasoning accuracy.
ReST-MCTS* D. Zhang et al. (2024)	LLM self-training	Uses process reward guidance with Monte Carlo Tree Search.	Produces higher-quality reasoning traces and improves models through iterative self-training.
InstructGPT Ouyang et al. (2022)	LLM alignment	Trains a reward model from human output rankings.	Improves instruction-following and alignment with human preferences.
LLM-Blender Jiang et al. (2023)	Candidate response ranking	Ranks and fuses outputs from multiple LLMs.	Shows that candidate ranking can improve generation quality over individual models.
Process supervision Lightman et al. (2023)	Mathematical reasoning	Trains reward models on intermediate reasoning steps.	Shows that step-level feedback can improve reasoning reliability compared with final-answer feedback alone.

response, adapting from memory, or using observed dialogue signals. Less attention has been given to explicitly comparing possible near-future client reactions before selecting a therapeutic response. This gap is important because MI-CBT dialogue depends heavily on timing: a response may appear supportive in the current turn but still increase resistance or weaken engagement in the next turn. Therefore, this thesis positions AIM-RGL as an application of future-aware, reward-guided response selection to simulated MI-CBT therapeutic dialogue, with the detailed framework and implementation described in Chapter 3.

CHAPTER 3

METHODOLOGY

CHAPTER 4

EXPERIMENT

CHAPTER 5

CONCLUSION

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APPENDICES

APPENDIX A

TITLE

Figure A1 Doge.



Note. Notice the different styling of figures in the appendix...